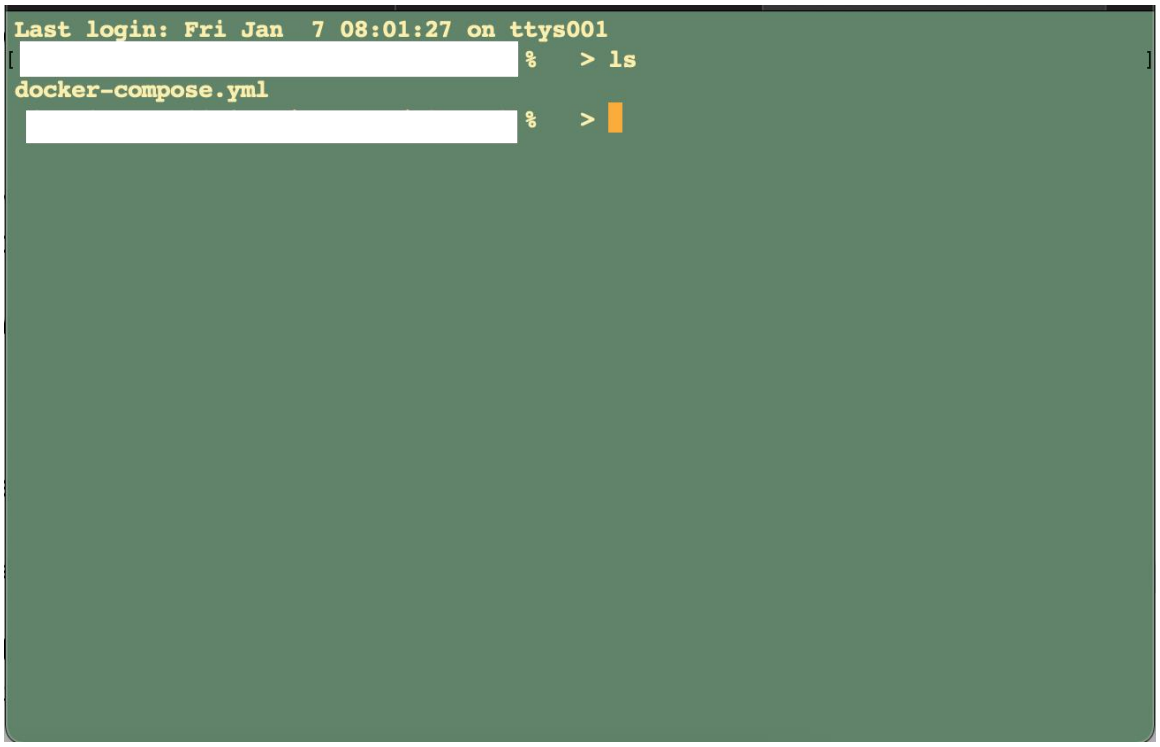
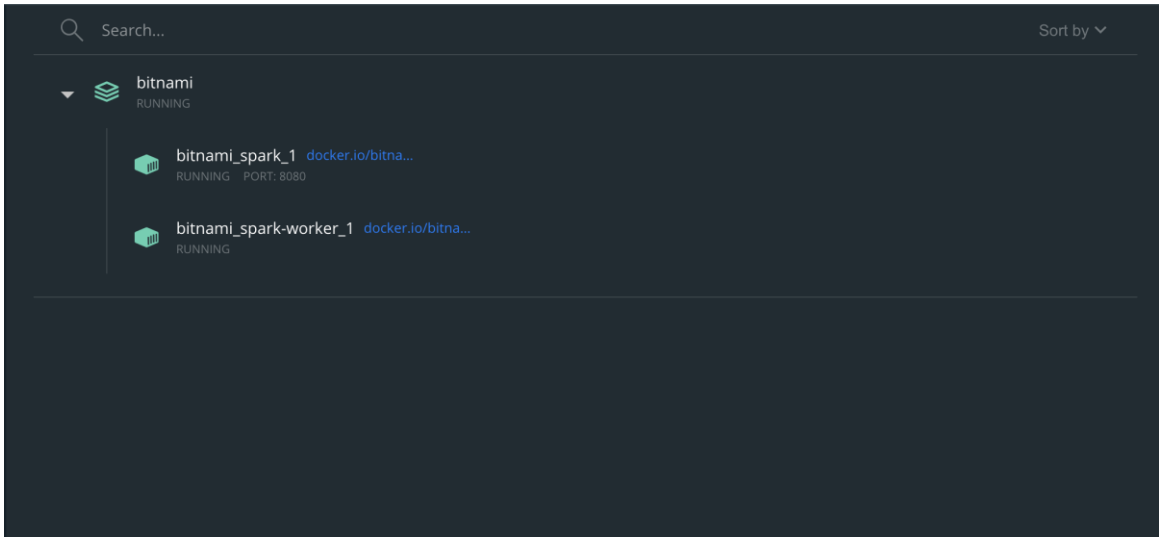
Final Assignment 19.1: Processing Big Data with Spark and Airflow

Part 1: PySpark

1.

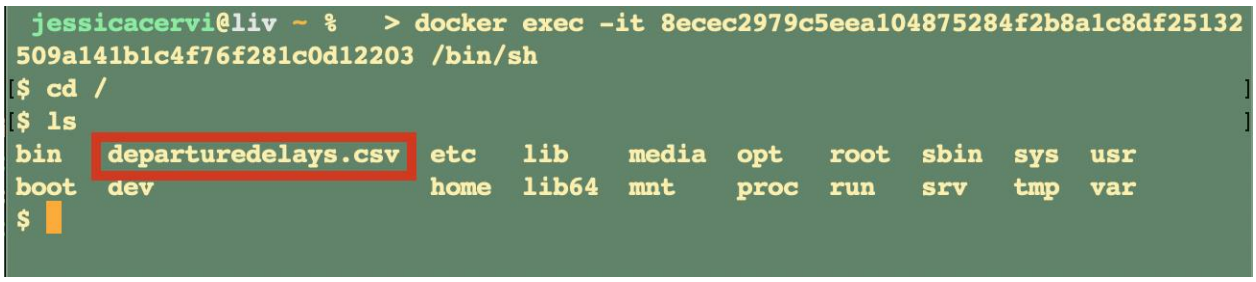


2.

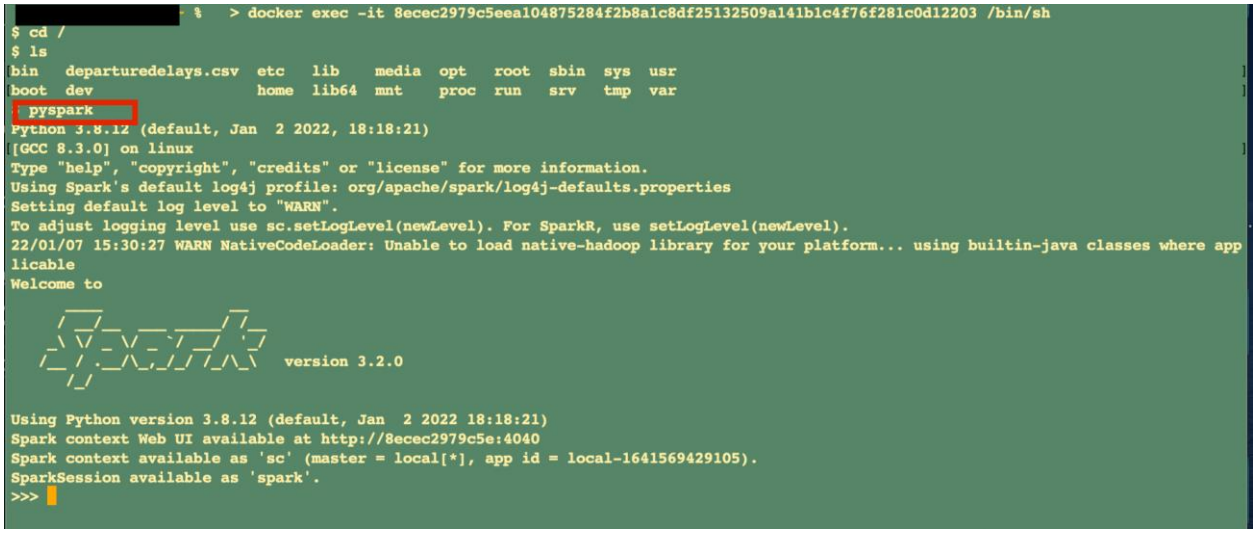


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3.



4.



5.



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6.

```
$ pyspark
Python 3.8.12 (default, Jan  2 2022, 18:18:21)
[GCC 8.3.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
Using Spark's default log4j profile: org/apache/spark/log4j-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
22/01/07 15:30:27 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where app
licable
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://8ecec2979c5e:4040
Spark context available as 'sc' (master = local[*], app id = local-1641569429105).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Assignment19.3")
...     .getOrCreate())
>>>
```

7.

```
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://8ecec2979c5e:4040
Spark context available as 'sc' (master = local[*], app id = local-1641569429105).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Assignment19.3")
...     .getOrCreate())
>>> assignment19_3_data = "/departuredelays.csv"
>>>
```

8.

```
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://8ecec2979c5e:4040
Spark context available as 'sc' (master = local[*], app id = local-1641569429105).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Assignment19.3")
...     .getOrCreate())
>>> assignment19_3_data = "/departuredelays.csv"
>>>
>>> df = (spark.read.format("csv")
...     .option("inferSchema", "true")
...     .option("header", "true")
...     .load(assignment19_3_data))
>>>
```

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9.

```
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 3.2.0
      /_/

Using Python version 3.8.12 (default, Jan  2 2022 18:18:21)
Spark context Web UI available at http://8ecec2979c5e:4040
Spark context available as 'sc' (master = local[*], app id = local-1641569429105).
SparkSession available as 'spark'.
>>> from pyspark.sql import SparkSession
>>> spark = (SparkSession
...     .builder
...     .appName("Assignment19.3")
...     .getOrCreate())
>>> assignment19_3_data = "/departuredelays.csv"
>>>
>>> df = (spark.read.format("csv")
...     .option("inferSchema", "true")
...     .option("header", "true")
...     .load(assignment19_3_data))
>>>
>>>
>>> df.createOrReplaceTempView("assignment19_3_table")
```

10.

```
>>>
>>> spark.sql("""SELECT date, delay, origin, destination
...     FROM assignment19_3_table
...     WHERE delay > 150 AND ORIGIN = 'PHL' AND DESTINATION = 'DFW'
...     ORDER by delay DESC""").show(15)
+-------+-----+------+-----------+
|   date|delay|origin|destination|
+-------+-----+------+-----------+
|1141620| 1177|   PHL|        DFW|
|2200820|  741|   PHL|        DFW|
|2141850|  316|   PHL|        DFW|
|1022039|  295|   PHL|        DFW|
|3131830|  280|   PHL|        DFW|
|1011425|  279|   PHL|        DFW|
|1110820|  277|   PHL|        DFW|
|3171140|  249|   PHL|        DFW|
|2231425|  242|   PHL|        DFW|
|3281830|  222|   PHL|        DFW|
|2190933|  199|   PHL|        DFW|
|2211425|  198|   PHL|        DFW|
|1081855|  193|   PHL|        DFW|
|2211620|  192|   PHL|        DFW|
|3251425|  182|   PHL|        DFW|
+-------+-----+------+-----------+
only showing top 15 rows
```

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11.

```
>>> spark.sql("""SELECT date, delay, distance, origin, destination
...     FROM assignment19_3_table
...     WHERE distance < 200
...     ORDER by distance DESC""").show(10)
+-------+-----+--------+------+-----------+
|   date|delay|distance|origin|destination|
+-------+-----+--------+------+-----------+
|3061235|  -16|     199|   SJT|        DFW|
|3161235|   -1|     199|   SJT|        DFW|
|3071235|   -8|     199|   SJT|        DFW|
|3081235|    6|     199|   SJT|        DFW|
|3091235|   31|     199|   SJT|        DFW|
|3101235|   -3|     199|   SJT|        DFW|
|3111235|    4|     199|   SJT|        DFW|
|3121235|   -2|     199|   SJT|        DFW|
|3131235|  -11|     199|   SJT|        DFW|
|3141235|   -7|     199|   SJT|        DFW|
+-------+-----+--------+------+-----------+
only showing top 10 rows
```

12.

```
>>> spark.sql("""SELECT date, delay, distance, origin, destination
...     FROM assignment19_3_table
...     WHERE distance >600
...     ORDER by distance DESC""").show(10)
+-------+-----+--------+------+-----------+
|   date|delay|distance|origin|destination|
+-------+-----+--------+------+-----------+
|1091625|    0|    4330|   HNL|        JFK|
|1051625|   18|    4330|   HNL|        JFK|
|1081625|   -5|    4330|   HNL|        JFK|
|1021625|  110|    4330|   HNL|        JFK|
|1041625|   -7|    4330|   HNL|        JFK|
|1061625|    0|    4330|   HNL|        JFK|
|1071625|    0|    4330|   HNL|        JFK|
|1011625|   -4|    4330|   HNL|        JFK|
|1111625|  -10|    4330|   HNL|        JFK|
|1031625|   -1|    4330|   HNL|        JFK|
+-------+-----+--------+------+-----------+
only showing top 10 rows
```

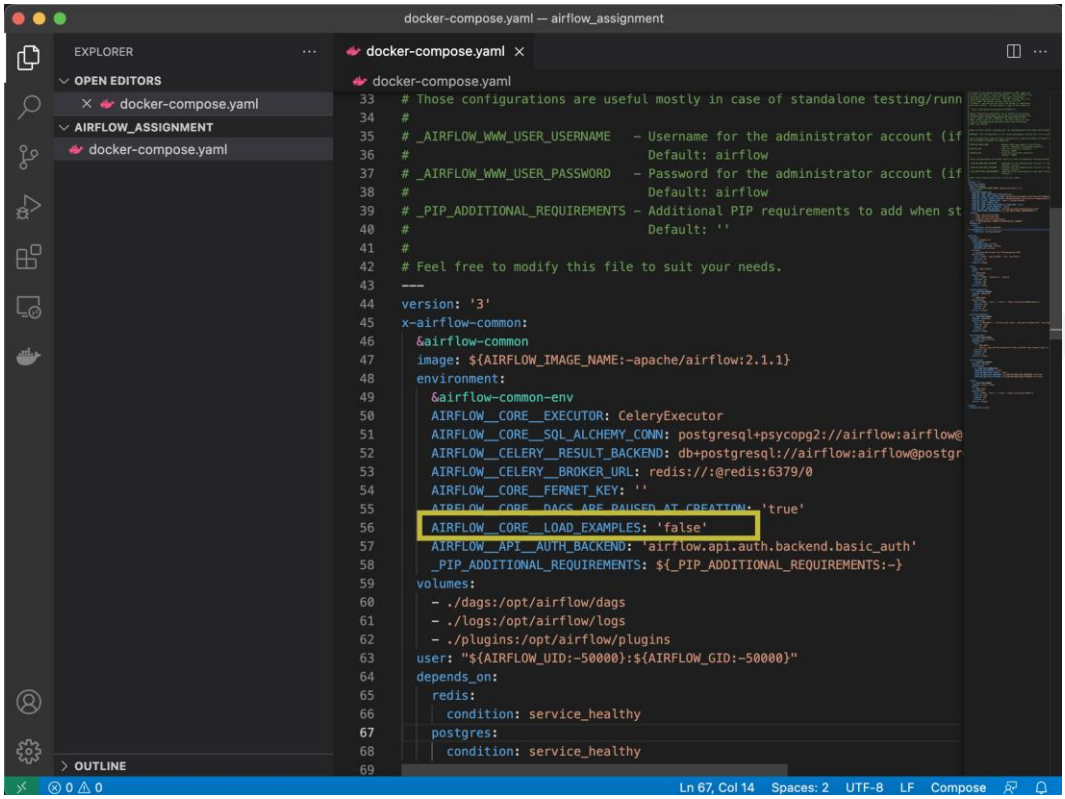
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Part 2: Airflow

1.

```
[redacted]~/Desktop/airflow_assignment %   > curl 'https://airflow.apache.org/docs/apache-airflow/2.1.1/docker-compose.yaml' -o docker-com
pose.yaml
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  4990  100  4990    0     0  13938      0 --:--:-- --:--:-- --:--:-- 13938
[redacted]~/Desktop/airflow_assignment %   >
```

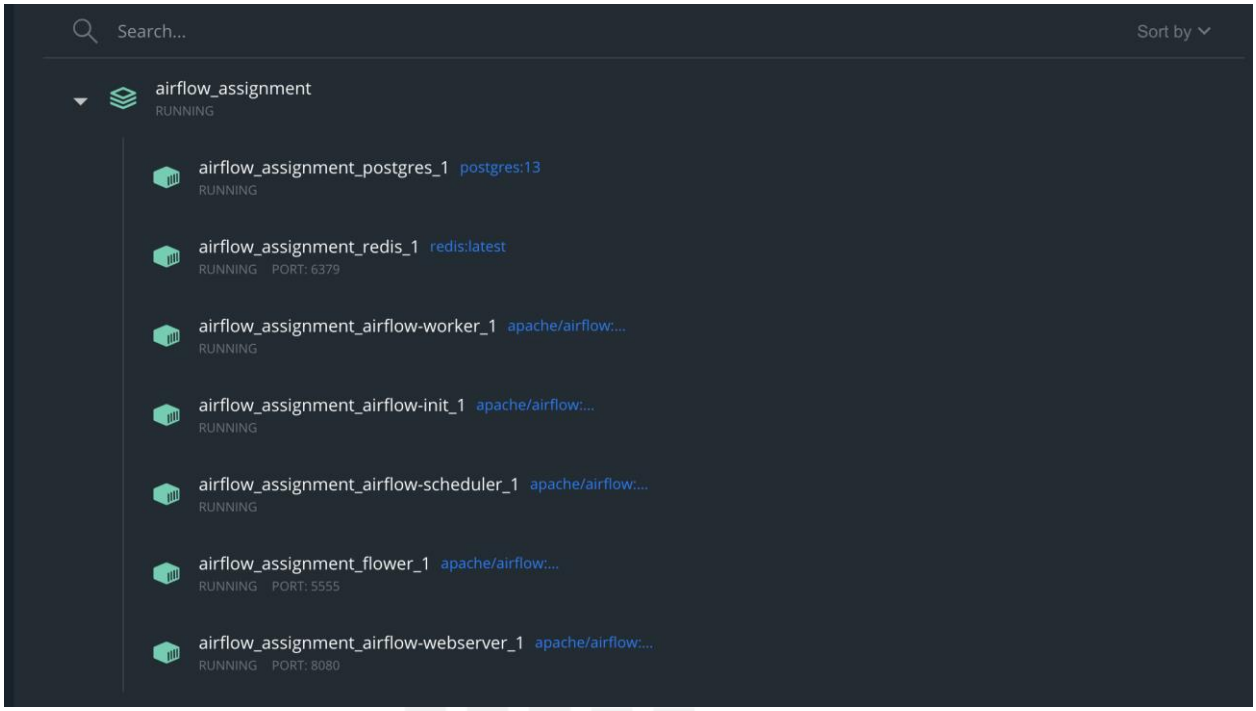
2.



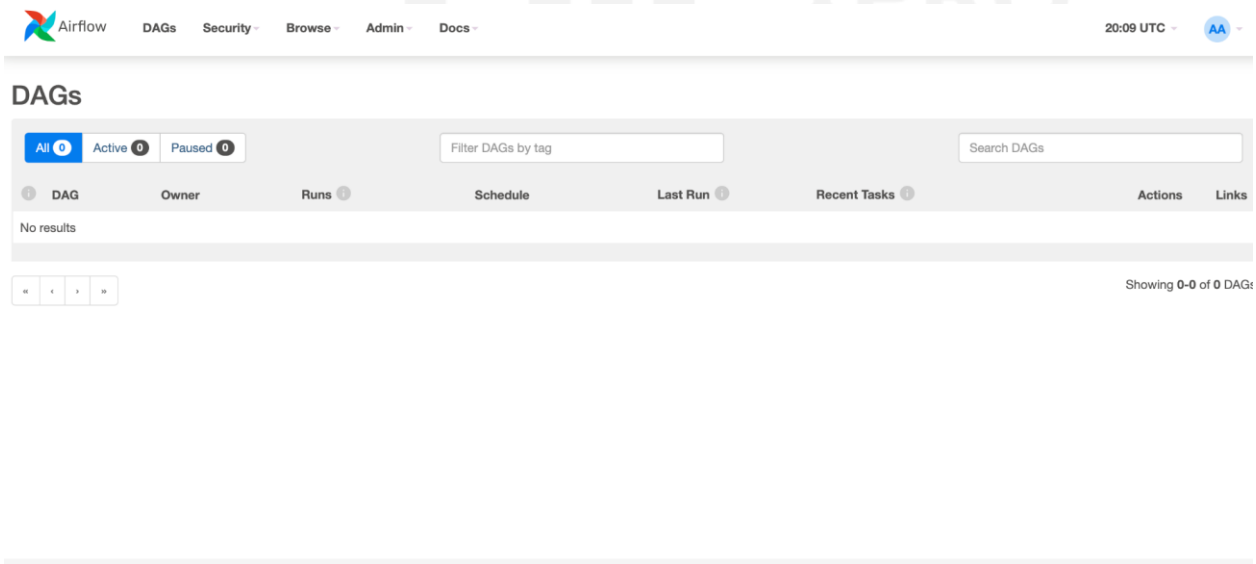

```yaml
# Those configurations are useful mostly in case of standalone testing/runn
#
# _AIRFLOW_WWW_USER_USERNAME   - Username for the administrator account (if
#                                Default: airflow
# _AIRFLOW_WWW_USER_PASSWORD   - Password for the administrator account (if
#                                Default: airflow
# _PIP_ADDITIONAL_REQUIREMENTS - Additional PIP requirements to add when st
#                                Default: ''
#
# Feel free to modify this file to suit your needs.
---
version: '3'
x-airflow-common:
  &airflow-common
  image: ${AIRFLOW_IMAGE_NAME:-apache/airflow:2.1.1}
  environment:
    &airflow-common-env
    AIRFLOW__CORE__EXECUTOR: CeleryExecutor
    AIRFLOW__CORE__SQL_ALCHEMY_CONN: postgresql+psycopg2://airflow:airflow@
    AIRFLOW__CELERY__RESULT_BACKEND: db+postgresql://airflow:airflow@postgr
    AIRFLOW__CELERY__BROKER_URL: redis://:@redis:6379/0
    AIRFLOW__CORE__FERNET_KEY: ''
    AIRFLOW__CORE__DAGS_ARE_PAUSED_AT_CREATION: 'true'
    AIRFLOW__CORE__LOAD_EXAMPLES: 'false'
    AIRFLOW__API__AUTH_BACKEND: 'airflow.api.auth.backend.basic_auth'
    _PIP_ADDITIONAL_REQUIREMENTS: ${_PIP_ADDITIONAL_REQUIREMENTS:-}
  volumes:
    - ./dags:/opt/airflow/dags
    - ./logs:/opt/airflow/logs
    - ./plugins:/opt/airflow/plugins
  user: "${AIRFLOW_UID:-50000}:${AIRFLOW_GID:-50000}"
  depends_on:
    redis:
      condition: service_healthy
    postgres:
      condition: service_healthy
```

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3.

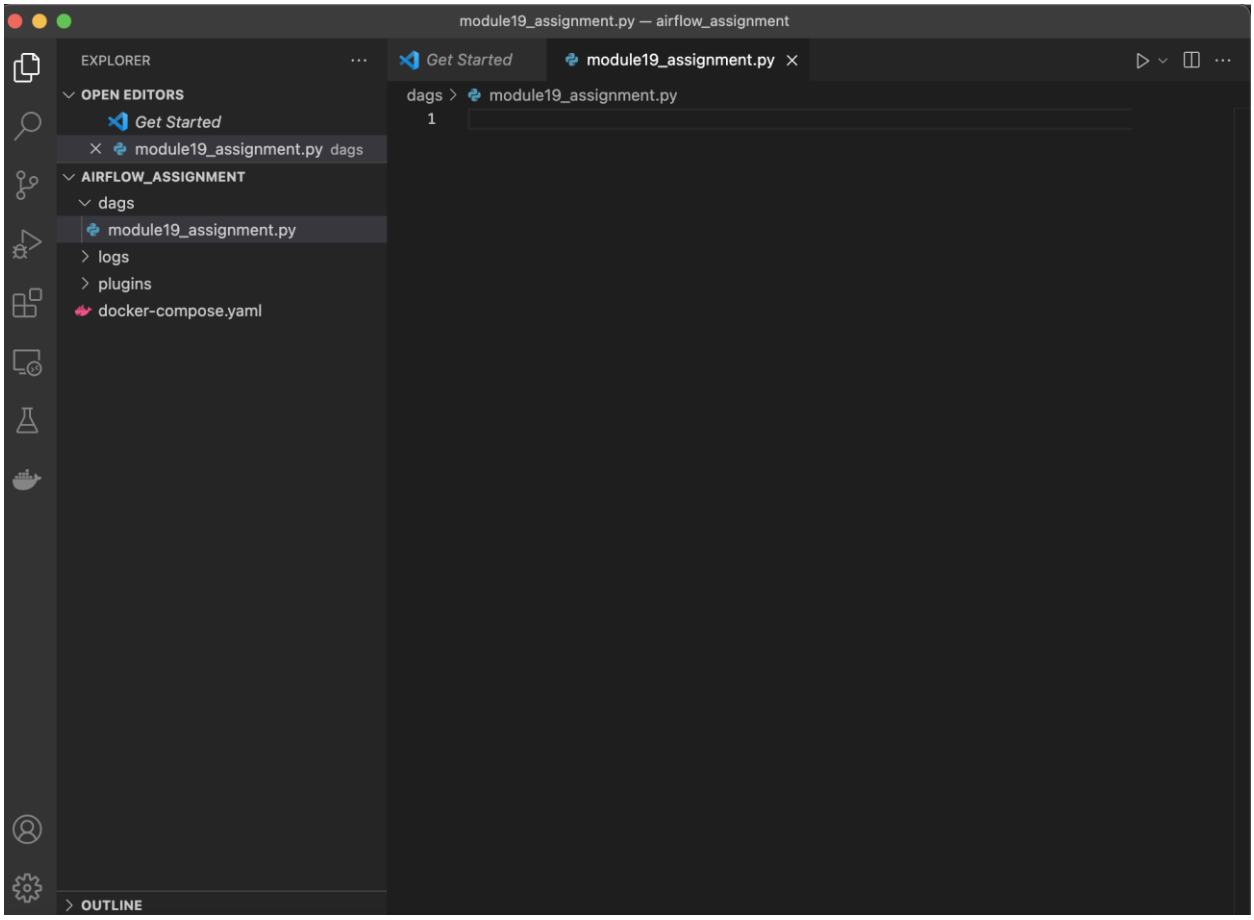


4.

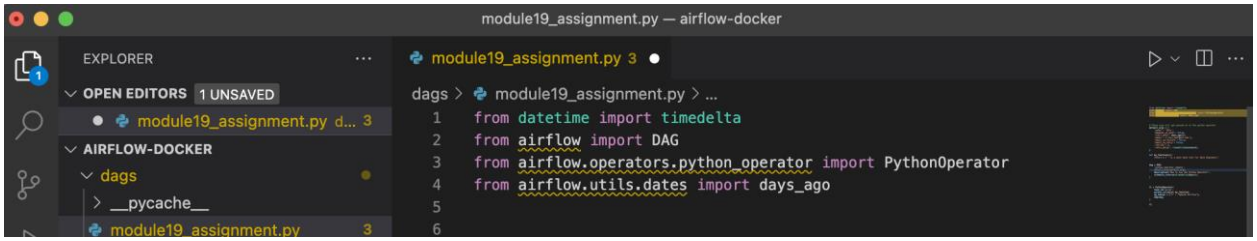


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5.

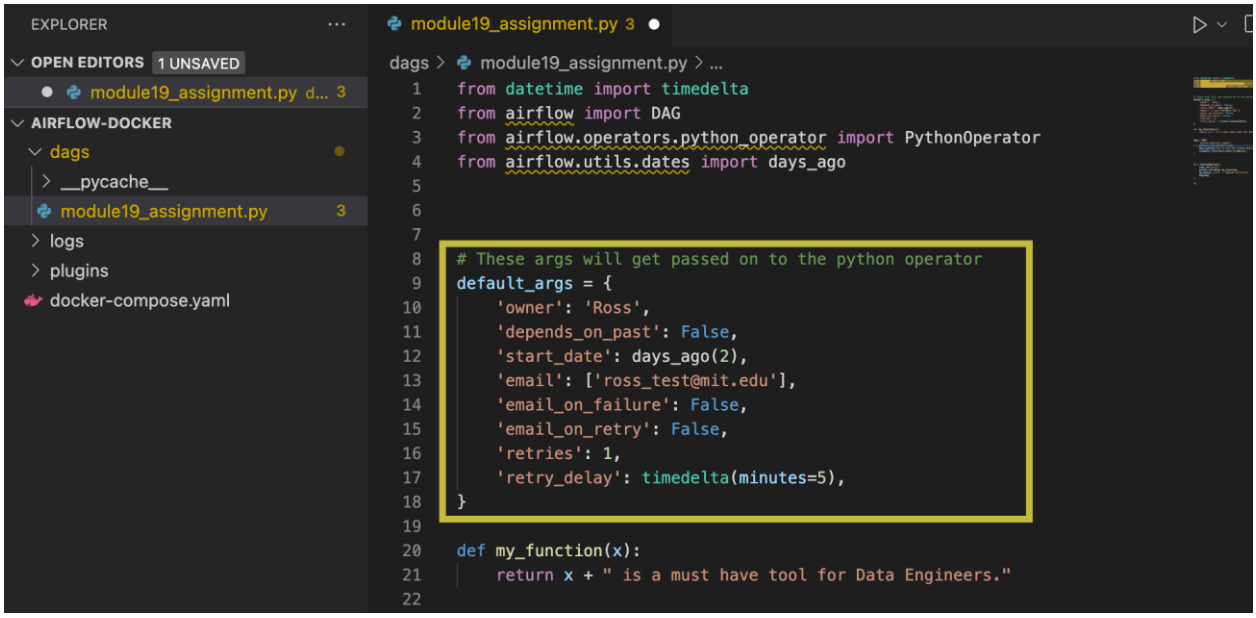


6.

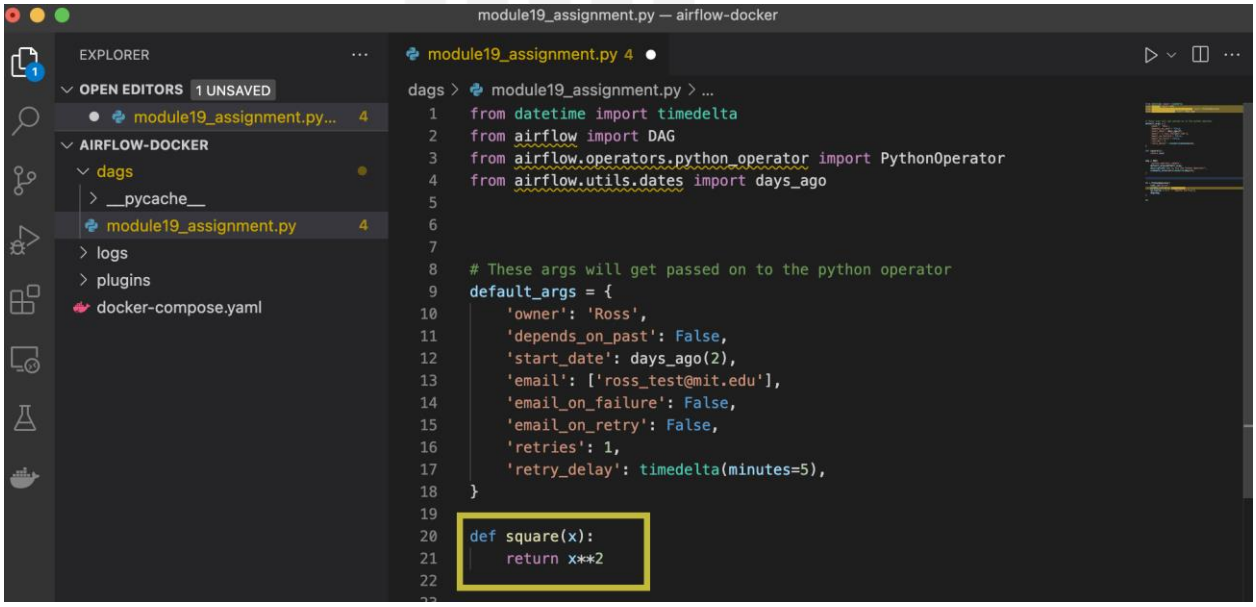


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7.



8.



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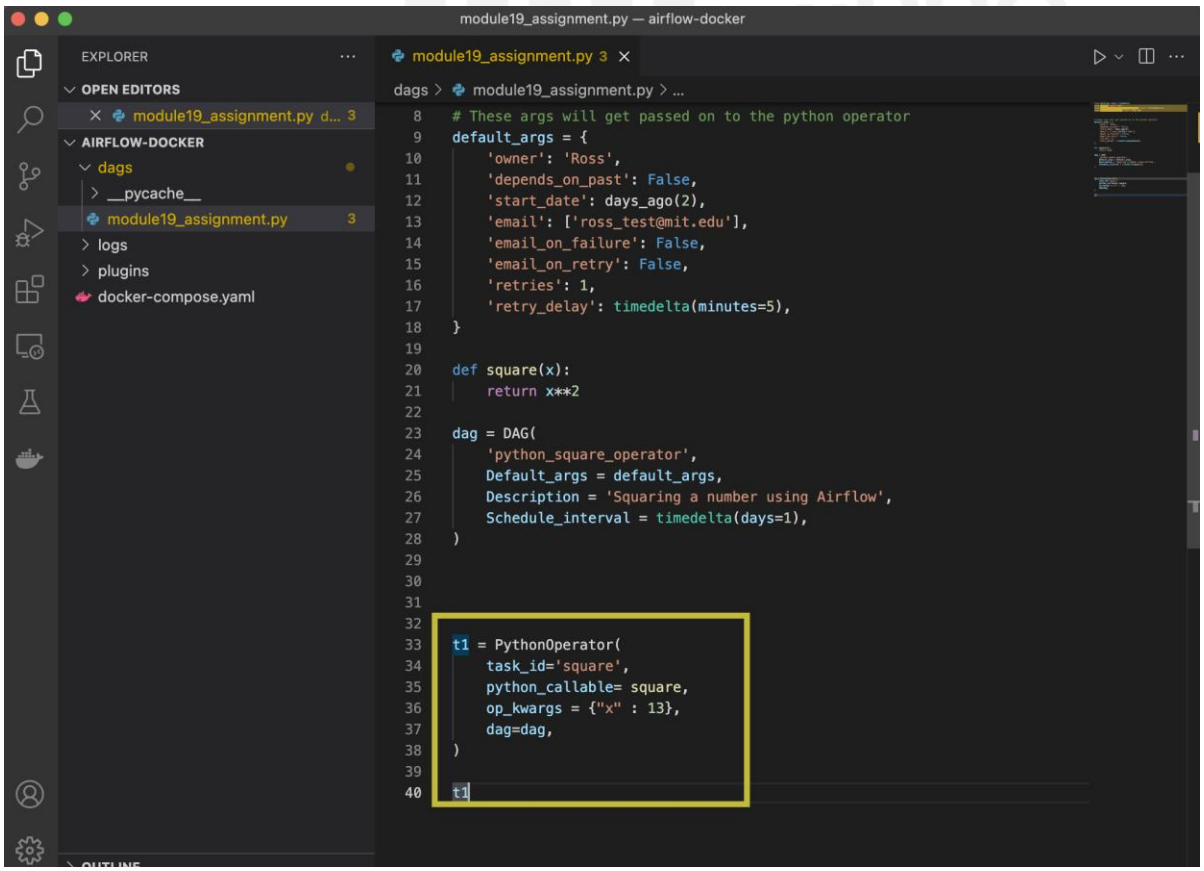
9.

```
# These args will get passed on to the python operator
default_args = {
    'owner': 'Ross',
    'depends_on_past': False,
    'start_date': days_ago(2),
    'email': ['ross_test@mit.edu'],
    'email_on_failure': False,
    'email_on_retry': False,
    'retries': 1,
    'retry_delay': timedelta(minutes=5)
}

def square(x):
    return x**2

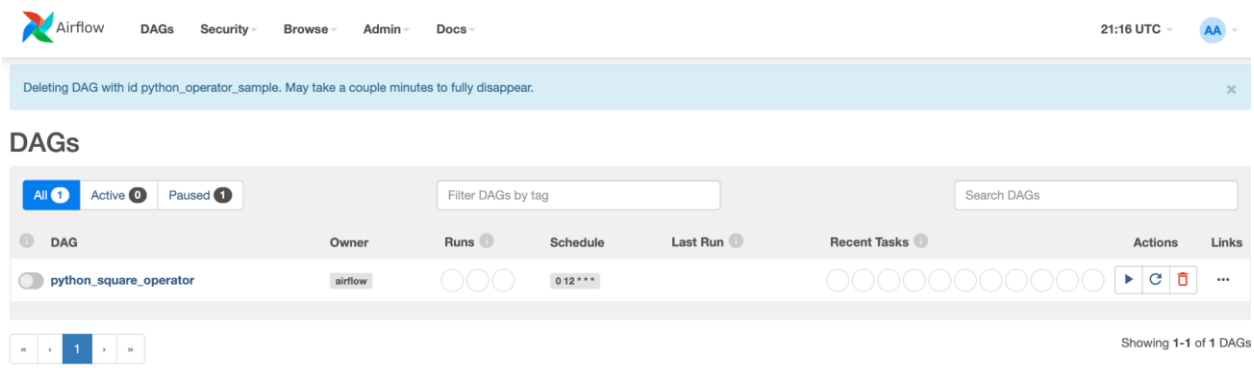
dag = DAG(
    'python_square_operator',
    description = 'Squaring a number using Airflow',
    schedule_interval='0 12 * * *',
    start_date=datetime(2017, 3, 20), catchup=False)
```

10.



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11.



12.



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